Big picture of Arrow Debreu Economy

1. In Arrow Debreu economy, every agent receive endowments. These endowments are given. Each agent knows the specific amount of endowments today and in each state in next period. What is uncertain is which state will realize in the next period.
2. The only things that an investor does is buying/selling different states' contingent claims from/to each other. They do it only **today**!
    a. In Arrow Debreu model, **every security decision is made today**, the only thing that agents do in the next period is to consume according to the number of state contingent claims at hand if a certain state realize and die… They don't make any decision in the next period.
    b. **They trade today because they are not satisfactory with the specific initial allocations of endowments. Under concave utility function, they want to smooth consumption across time and states.** For example, they don't want to have zero consumption for a state, otherwise, if that state realizes in the next period, they'll become miserable.

3. The price of a state contingent claim i denotes how many units of current consumption one state contingent claim i can be used to exchange for. This equation and its interpretation is key. The pricing equation is derived from each investor's utility optimization problem.
    a. $q^i = \frac{\beta \pi_i U'(C_j^i)}{U'(C_j^0)}$, this equation links security prices to consumption, i.e. economic fundamentals
    b. The more patient investors are, the higher the weight the utility from consumption in the next period, i.e. the higher the β, the more valuable the state contingent claims that deliver goods in the next period.
    c. **State contingent claim i is risky!** It deliver one unit of payoff in the next period if and only if state i realizes, and it'll become useless in the next period if other states realize. So the price of state contingent claim depends on the likelihood of the corresponding state i, the higher the likelihood, the more valuable the price.
    d. The higher the C_j^i, that is, the better(worse) the economic condition in the next period state i, the lower (higher) the less (more) valuable the state contingent claim i which has one unit of payoff in state i.
    e. Expected gross returns E(R^i)=E($r^i + 1$)= $\pi_i * 1/q^i$. The numerator is the expected payoff of state contingent claim i, the denominator is today's price for state contingent claim i. So the higher the q^i, the lower the E(R^i) or E(r^i). Combined d and e, state contingent claim i that deliver one unit of payoff in good (bad) state, or when C_j^i is high (low) => has low (high) q^i => high (low) expected returns E(R^i) or E(r^i)

4. Markets in Arrow Debreu economy are competitive, given concave utility function, the economy is also Pareto optimal=>It's equivalent to solving Arrow Debreu Economy VS solving social planner's problem
5. From solving social planner's problem, we realize that when there is no risk in aggregate endoments ,i.e. the same aggregate endowment across states=> there is no risk in individual investor i's consumption, i.e. the same individual consumption across states.
    a. Because solution in Arrow Deberu economy is the same as that in social planner's problem, the same conclusion applies to Arrow Deberu economy

   b. Point a above implies that even though the initial individual endowments might be risky, i.e. different endowments across states for each investor, as long as the aggregate endowments is constant across states (i.e. no risk in aggregate endowment), the individual consumption is constant across states (i.e. no risk in individual consumption)
   c. From point b above, trading in Arrow Debreu indeed helps investors to smooth consumption across time and states!
6. State contingent claims are imaginary basic securities. This security has simple payoff, so helps us to theorize how to link consumptions to security prices. “complex securities” like stocks can be thought as a bundle of state contingent claim prices.